Circle-tracing and serial subtraction task - Insight46 phase 1

Kirsty Lu | 24/07/2019

Document details

Categories of variables	Circle-tracing and serial subtraction task
Summary of work undertaken	1. Data collected, processed and cleaned as described below. 2. Summary outcomes calculated as described below.
Names of source variables	id, (participant's, ID, number), Ccomp, Cwn, Cdual, Cdualwn, Csingle_comp, Cnumrot, Cout_numdev, Cin_numdev, Cnumdev, Cnumdev_rot, Cout_away_t, Cout_toward_t, Cin_away_t, Cin_toward_t, Cann_t, Cann_l, Cout_away_l, Cout_toward_l, Cin_away_l, Cin_toward_l, Cr, Ctheta, Csumpos, Csumneg, Ctot, Ccorr
Output variables	circle_comp_i46p1, circle_wn_i46p1, circle_rateo_i46p1, circle_dual_i46p1, circle_single_comp_i46p1, circle_numrotd_i46p1, circle_numdev_rotd_i46p1, circle_toto_i46p1, circle_errn_i46p1, circle_erro_i46p1, circle_dualwn_i46p1
Papers using these variables	Characterising cognition in later life and its relationship with biomarkers of Alzheimer's disease: a study of members of the National Survey of Health and Development (the British 1946 birth cohort). PhD thesis (Kirsty Lu) **It is recommended that potential users of these datasets discuss their analyses with Kirsty Lu.**

Background

A brief description has been published in the Insight 46 protocol paper:

1 Lane et al. (2017) Study protocol: Insight 46 – a neuroscience sub-study of the MRC National Survey of Health and Development, BMC Neurology, 17:75, doi: 10.1186/s12883-017-0846-x

The circle-tracing task was presented on a Lenovo ThinkPad X61 tablet laptop, placed horizontally on the table in front of the participant, with an additional vertically placed monitor behind it for the indirect condition. The monitor was at a distance of about 60cm from the participant. On the laptop, the display showed a 90mm diameter circle and 5mm thick annulus; on the monitor, the display showed a 143mm diameter circle and 9mm thick annulus. Although the actual size of the circle was larger on the monitor than the laptop, the visual angle was comparable – about 13o in each case – because the monitor was at a greater distance from the eyes.

Participants were instructed to trace clockwise round the circle using a stylus as quickly and accurately as possible (trying to stay within the annulus) without leaning their hand on the screen, starting from the vertical apex of the circle. A thin blue line appeared on the display to show their tracing path. In the direct condition, participants could see their hand and the path they were tracing on the tablet screen. In the indirect condition, the laptop was covered by an upturned box with the front open to allow the participant to put their hand inside. The participant wore a long cape which completely covered both their arms so they had no direct visual feedback while they were drawing, but they could view a copy of the circle and the tracing path on the monitor.

The length of each trial was 45 seconds. There were three trials of each feedback condition, administered in the order Direct, Indirect, Indirect, Direct, Direct, Indirect. Two practice trials – one direct and one indirect – were administered approximately one hour before the main experiment, so that participants could familiarise themselves with the procedure, but with a long enough delay to mitigate against immediate practice effects.

The task was administered in a dual-task format with a concurrent task of serial subtraction. Participants were asked to count backwards in threes from a given starting number while performing each circle-tracing trial. Starting numbers across the six trials were the same for all participants as follows: 99, 98, 97, 96, 95 and 94. Participants were instructed that, if they reached zero or near zero, they should begin again from the starting number. The tester wrote down the sequence of numbers called out by the participant.

In December 2016 (approximately half-way through the data collection period) the procedure was modified to add an additional two circle-tracing trials (Direct, Indirect) to the end of the experiment. These two trials were performed as a single task without serial subtraction, to allow comparison of circle-tracing performance in the dual and single task conditions in a sub-sample of participants, to investigate how tracing speed and accuracy were influenced by having to attend to the concurrent subtraction task (the “dual-task cost”).

Data processing and cleaning

Out of 502 Insight46 participants, 487 were recorded in XNAT as having completed the dual-task circle-tracing (Ccomp). However, 4 of these did not have useable data. Two new variables were created: one to indicate whether or not participants had useable dual-task circle-tracing data (Cdual) and one to record the reason why data were not usable (if applicable – Cdualwn). 209 participants were administered the single-task circle-tracing (Csingle_comp).

Each participant’s raw data file was processed using the HD-CAB Data Analyzer software application PxAnalyze, which outputted a single spreadsheet of outcome variables for all participants. This spreadsheet was imported into Stata, and transformed into a format whereby each participant had one row per trial.

Trials were labelled with their number (Ctrial) (1 – 6 for the dual task, 7 – 8 for the single task), condition (Ccon – direct or indirect) and task type (Ctask – dual or single).

For each trial, the following outcome measures were generated (variable names in red):

1. Number of rotations completed in 45 seconds (Cnumrot), as an index of tracing speed. This was recorded by the software to 14 decimal places.
2. An error was recorded whenever the stylus deviated outside of the annulus, either beyond its inner edge (Cin_numdev) or outer edge (Cout_numdev) for more than 100 ms. These variables were summed to create a total number of errors (Cnumdev), which was expressed per rotation, as an index of tracing accuracy (Cnumdev_rot), by dividing Cnumdev / Cnumrot.
3. Error time was recorded while the stylus was tracing outside the annulus. Error time was separated according to whether the stylus deviated beyond the inner or outer edge of the annulus, and whether it was moving away or towards the annulus (Cin_away_t, Cout_away_t, Cin_toward_t, Cout_toward_t). Total time moving away from the annulus (Caway_t, considered to represent error detection time) was calculated by summing Cout_away_t + Cin_away_t. This was expressed per rotation (Caway_t_rot) by dividing Caway_t / Cnumrot. Total time moving back towards from the annulus (Ctoward_t, considered to represent error correction time) was calculated by summing Cout_toward_t + Cin_toward_t. This was expressed per rotation (Ctoward_t_rot) by dividing Ctoward_t / Cnumrot.
4. A ‘tracing time’ variable (Cttot) was calculated by summing the total time spent tracing inside the circle (Cann_t) and the total time spent tracing outside the circle (Cout_away_t + Cout_toward_t + Cin_away_t + Cin_toward_t)
5. The total length of the line traced inside the circle (Cann_l).
6. The total length of the line traced outside the circle (i.e. when making errors), split according to whether the stylus deviated beyond the inner or outer edge of the annulus, and whether it was moving away or towards the annulus (Cin_away_l, Cout_away_l, Cin_toward_l, Cout_toward_l).
7. Mean radius over all the sampled points (Cr)
8. Mean theta over all the sampled points (Ctheta)
9. Two additional variables are included present, labelled as “Sum Positive Theta Deltas” (Csumpos) and “Sum Negative Theta Deltas” (Csumneg). I corresponded with the research group in Monash University who wrote the software and they did not have a definition of these variables, as the author of the source code was not available.

Certain trials were excluded from analysis according to the following criteria:

1. The total tracing time (Cttot) should be 45 seconds (i.e. the duration of the trial) but could be less if the trial was not completed perfectly, for example if the participant lifted the pen off the screen during the trial, they

pressed too hard with the stylus such that no trace was recorded temporarily, or they failed to begin tracing immediately upon initiating the trial. The maximum time in this dataset was exactly 44 seconds rather than 45 seconds as expected. I corresponded with the research group in Monash University who wrote the software and they did not have an explanation for this discrepancy, as the author of the source code was not available. If the total tracing time was less than three standard deviations below the mean (34.7 seconds), the trial was excluded (i.e. all outcome variables were changed to 'missing' except Cttot, which was left intact for reference). This resulted in thirty-two participants having one trial excluded for being below this threshold, an additional three participants having two trials excluded, and an additional one participant having four trials excluded. After these exclusions, the number of participants with usable single-task data dropped to 208 and the number with usable dual-task data remained at 483.

2. An additional two participants had one indirect trial excluded because the output showed a negative number of rotations (-0.07 and -0.08). All outcome variables were changed to 'missing' for these trials, except Cttot which was left intact for reference). I corresponded with the research group in Monash University who wrote the software and they did not have an explanation for these odd results, as the author of the source code was not available.

For the serial subtraction data, trials were excluded where the concurrent circle-tracing data had been excluded, as these exclusions were mostly due to the tracing time being significantly less than expected, indicating a problem with performing the circle-tracing. It was observed that when such problems arose, the participant invariably paused their subtraction while trying to address it. This resulted in 34 participants having subtraction data excluded for one trial and an additional two participants having two trials excluded. The available subtraction data were number of correct responses (Ccorr) and total number of responses (Ctot). The subtraction rate (Crate, in responses per second) was calculated by dividing Ctot by 45. The number of incorrect responses (Cincorrect) was calculated by subtracting Ctot-Ccorr, and then converted into the percentage (Cerr) as: Cincorrect/Ctot*100.

This dataset was saved as Insight46_circletracing_fulldata_cleaned_final_20190724 Note that it contains only the 483 participants with useable data from the dual-task circle-tracing and serial subtraction task. For details of participants who did not have useable data – and reasons why – see the summary dataset described below.

A second dataset was created which contains one row per participant, summarising their performance across the dual-task trials of the experiment. The following variables were calculated: 1. Mean number of rotations (Cnumrotd), calculated by averaging Cnumrot. I did this by first calculating the mean of Cnumrot for the direct and indirect conditions separately, then calculating the mean of these. This was preferable to simply averaging across all the trials, in order to avoid an unfair outcome for the participants who had one or more trials excluded. For example, if a participant was missing an indirect trial, a simple average of their scores would be weighed towards direct trials (known to be easier) and the average score would look better than it should.

1. Mean number of errors per rotation (Cnumdev_rotd), calculated by averaging Cnumdev_rot. As above, I did this by first calculating the mean of Cnumdev_rot for the direct and indirect conditions separately, then calculating the mean of these.
2. Mean subtraction rate (Crateo), calculating by averaging Crate across all trials
3. In order to calculate the percentage of incorrect subtraction responses (Cerro), it was preferable not to simply average Cerr, as this would mean that errors committed on trials where subtraction was slower would be weighted more heavily than errors committed on trials where subtraction was faster. To avoid this, the total number of incorrect responses summed across all trials was first calculated (Cerrn), and the total number of responses summed across all trials trials (Ctoto). The overall percentage of incorrect subtraction responses was then calculated as: Cerrn/Ctoto*100

This dataset was saved as Insight46_circletracing_summaryoutcomes_cleaned_final_20190724 Note that it contains all 502 Insight46 participants.

Variables in this document

9 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Visuomotor performance [60427]		
circle_comp_i46p1	Circle tracing: was the task completed? - Insight46 phase 1	topsheet field
circle_dual_i46p1	Circle tracing: participant has usable dual-task data - Insight46 phase 1	topsheet field
circle_errn_i46p1	Circle tracing: total number of incorrect subtraction responses summed across all 6 trials - Insight46 phase 1	topsheet field
circle_erro_i46p1	Circle tracing: percentage of incorrect subtraction responses summed across all 6 trials combined - Insight46 phase 1	topsheet field
circle_numdev_rotd_i46p1	Circle tracing: mean number of errors per rotation across all dual-task trials - Insight46 phase 1	topsheet field
circle_numrotd_i46p1	Circle tracing: mean number of rotations across all dual-task trials - Insight46 phase 1	topsheet field
circle_rateo_i46p1	Circle tracing: mean subtraction rate (responses per second) - Insight46 phase 1	topsheet field
circle_single_comp_i46p1	Circle tracing: Participant completed the single-task circle-tracing trials - Insight46 phase 1	topsheet field
circle_toto_i46p1	Circle tracing: total number of subtraction responses summed across all 6 trials - Insight46 phase 1	topsheet field

1 medium-confidence match

Variable	Label	How it was found
Clinical data [51] › Physical measures [110] › Hand examination [1106]		
problems	Any problems identifed with the hands - Hands Examinations 1999	prose scan

2 low-confidence matches

Variable	Label	How it was found
Clinical data [51] › Physical measures [110] › Hand examination [1106]		
hand	Now I would like to examine your hands for any bumps or swellings. Would you be willing for me to examine your hands? - first record - at age 53 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Musculoskeletal system conditions [51016]		
back	In the last 12 months, have you had sciatica, lumbago or severe backache at age 53 years	prose scan